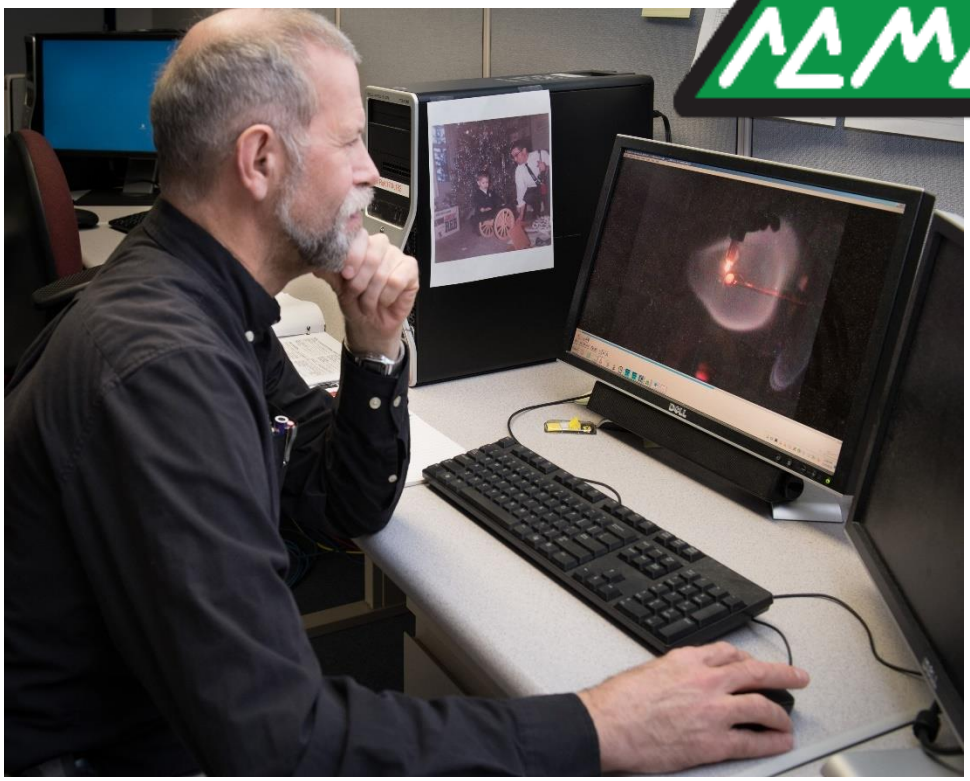
Advanced Combustion via Microgravity Experiments (ACME)
Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames)

ACME/E-FIELD Flames Data Interpretation

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Dennis Stocker, the ACME Project Scientist, at the science console for the Combustion Integrated Rack (CIR) in NASA's Glenn ISS Payload Operations Center (GIPOC). In this photo, he is reviewing data from an ACME experiment other than E-FIELD Flames.

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Table of Contents

	Page
1. Introduction	**2**
2. E-FIELD Flames Overview	**3**
3. Tests Conducted	**4**
4. Data Interpretation	**6**
A. Non-Imaging Data	**6**
1. Data Output File	6
2. Mass Flow Controller (MFC) Setpoint File	9
3. Fuel Oxidizer Management Assembly (FOMA) data	10
4. Space Acceleration Measurement System (SAMS) data	11
B. Image Files	**12**
1. Analog, operations (ops), internal camera	12
2. ACME Data Camera (DataCam)	13
3. Cool Flames Investigation (CFI) intensified camera	19
4. High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package	23
C. Test Parameters and Other Files	**24**
1. Test parameter file	24
2. Notes	24
3. Test timing	25
4. Logs	26
Appendices	
A. Nomenclature	**37**
B. Installation of QtOMA2 on a PC	**38**

1. Introduction

The purpose of this report is to describe the data acquired for ACME's E-FIELD Flames experiment to facilitate its analysis. The data is all digital, where the fuel was gaseous and there was no collection of soot or the combustion gases. For each test, the key data consists of image files and a file with the non-imaging measurements. Additional files such as camera metadata and operational logs provide information that can support the analysis.

It is recommended that readers interested in analyzing the E-FIELD Flames data begin by familiarizing themselves with the hardware and instrumentation, either through the overview in the E-FIELD Flames Final Report or the more comprehensive description found in the ACME Engineering Implementation document.

Those seeking to analyze the data should next review this document. To interpret the non-imaging data, it is suggested that a sample data file be opened, named for example 18073A3_data.csv, which has comma-separated values and can be opened with spreadsheet software such as Microsoft Excel. The data is in raw voltages, where conversion to engineering units requires study of ACME's Software Interface Design Description, i.e., ACME-DOC-010 (where the latest version at the time of this writing is revision D).

Additional documents that may be important include:

ACME-RPT-010 A	Data Camera Characterization Report, Rev. A
ACME-RPT-011	Radiometer Characterization Report
ACME-RPT-013	PMT Characterization Report

However, the BRE Radiometer Characterization Report, ACME-RPT-012, is not relevant because the BRE radiometer was developed specifically for ACME's Burning Rate Emulator (BRE) experiment. E-FIELD Flames instead made use of the standard radiometers addressed in ACME-RPT-011.

2. E-FIELD Flames Overview

This text is the "Overview" from the ACME/E-FIELD Flames Final Report, which provides descriptions of the science, preliminary results, and more. It is suggested that it be reviewed by those interested in analyzing the E-FIELD Flames data.

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which tests are conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. Derek Dunn-Rankin of the University of California, Irvine (UC Irvine). Dr. Yu-Chien (Alice) Chien, also of UC Irvine, served as the Project Scientist at the university. Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is NASA's Project Scientist (PS) for ACME including E-FIELD Flames and the rest of the ACME investigations. In preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team includes Drs. Uday G. Hegde and Vedha Nayagam who are also with CWRU.

The experiment had two Co-Investigators, the first of which was Prof. Felix Weinberg of London's Imperial College. Prof. Weinberg was a pioneer in researching the effects of electrical fields on flames in microgravity, but unfortunately died in 2012 (see http://www.imperial.ac.uk/news/117808/tribute-felix-weinberg/) before the hardware was built and the tests were conducted on the ISS. The second Co-Investigator was Dr. Sunny Karnani who did his doctoral research in preparation of the spaceflight experiment and afterward worked for ClearSign Combustion Corp., which markets Electrodynamic Combustion Control (https://www.clearsign.com/technology/electrodynamic-combustion-control/). The corporation provided some support to the E-FIELD Flames research effort, but Karnani left ClearSign to conduct research for the U.S. Army, effectively ending his intimate role in the project prior to the spaceflight operations. He remains a collaborator.

E-FIELD Flames operations were conducted in two rounds of testing during Increments 55-57, which were more specifically in March-May and September-November 2018. The break in testing (from May to September) was intentional and planned years in advance to allow the research team to analyze early results and use the insights obtained from the analysis to plan the second round so as to maximize its scientific output.

3. Tests Conducted

This text is an excerpt from the "Results" section of the ACME/E-FIELD Flames – 30-Day Operation Accomplishments Report for ISS Increments 57 and 58 (which is included in the ACME/E-FIELD Flames Final Report).

A total of 219 flames were ignited for the E-FIELD Flames experiment, with 142 and 77 respectively ignited during Increments 55-56 and 57-58. Note that the investigation's testing was completed during Increment 57, so no flames were ignited during Increment 58. The first and second rounds of testing respectively occurred in March-May 2018 and September-November 2018. The two rounds of tests resulted in 120 and 99 ignited flames conducted with coflow and gas-jet burners, respectively. The coflow burner was developed for the Coflow Laminar Diffusion Flame (CLD Flame) experiment, where CLD Flames' second round occurred between the two rounds of the E-FIELD Flames experiment. The 'interruption' in the E-FIELD Flames testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

E-FIELD Flames does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH_4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C_2H_4), is highly sooting because of its carbon double bond. For both fuels, tests were conducted with pure fuel and fuel mixed with nitrogen over a broad range of dilutions. For limited cases, the fuel and nitrogen was premixed on the ground prior to launch. ACME bottles were launched with methane concentrations of 30%, 40%, 45%, and 100% with a balance of nitrogen (on a volume basis). Similarly, bottles were launched with ethylene concentrations of 30%, 50%, and 100%, again with a balance of nitrogen (on a volume basis). But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations. In all cases, the coflow was 21% oxygen with a balance of nitrogen (on a volume basis).

No samples (e.g., of soot) were collected for E-FIELD Flames; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the E-FIELD Flames ignited for E-FIELD Flames during Increments 55-57, including a limited overview of the test conditions is presented here.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55-57
(*where the investigation was completed prior to Increment 58*)

Test Day[1]	Year	Month	Date	Mesh Polarity[2]	Fuel bottle	Fuel conc.[3]	Flames ignited[4]
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Second round of tests, with the gas-jet burner, in Sept.-Nov. 2018 (Increments 56-57)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11
18282			9		100% CH4	70%, 100%	13
18285			12		100% C2H4	100%	10
18290			17			80%	11
18298			25	positive	40% CH4	40%	9
18302			29		30% C2H4	30%	10
18312		11	8		100% CH4	70%, 100%	13
18319			15		100% C2H4	25%, 50%, 80%, 100%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

4. Data Interpretation

This key section provides (1) a general description of the raw experimental data and associated files and (2) guidance on data conversions, e.g., to engineering units. Only limited guidance is provided for the processing of image data, with the important exception of the conversion of the ACME DataCam images from the raw Halcon object (HOBJ) format.

A. Non-Imaging Data

Data and files discussed in this section include the following:

Data Output File
Mass Flow Controller (MFC) Setpoint File
Fuel Oxidizer Management Assembly (FOMA) data
Space Acceleration Measurement System (SAMS) data

Some ACME experiments featured measurements made with the CIR's Gas Chromatograph Instrumentation Package (GCIP), but it was not used for the E-FIELD Flames experiment and a such that data type is not addressed in this document.

1. Data Output File

The basic ACME non-imaging measurements for each test are together in a file described in Section 8.1 Data Output File on page 124 of Software Interface Design Description (ACME-DOC-010 Rev D). The test number is made up of the year (2-digts), day of year (3 digits), and test (a letter and single-digit number). As such, the example, 18073A3 is test A3 conducted on the 73rd day of 2018. But the test could also be early on the next (e.g., 74th) day because the day portion of ACME test numbers is not changed when operations continue past midnight (whether GMT or Eastern) into the start of the next day.

Tests numbers that only differ by the final digit (e.g., 18073A1, 18073A2, and 18073A3) are normally similar and occasionally identical, where the number is indicative of the variant or repeat. Letters I and O are not used because of their similarity to 1 and 0. While initially used in ACME testing, the letter E ceased to be used because Excel software (for example) by default interpreted it as representing an exponential which was inconvenient. As a related aside, ACME tests are often not conducted in order and some planned tests are regularly not conducted at all. The latter results in additional letter gaps if the tests from a day are sorted alphabetically. However, ACME testing often begins with test A1, where the final digit begins with 1 and is increased from there with variant or repeat tests for the same letter.

The basic data is in a CSV format, i.e., with comma-separated values, which can be opened by spreadsheet software such as Microsoft Excel. The first row is a header and provides a limited summary of the hardware and operational conditions. The header is explained in Section 10.1 Metadata Header of ACME-DOC-010 Rev D, which begins on page 126.

As an example, the header from 18073A3_data.csv includes the following information (displayed in two columns rather than a long row):

2018-03-14 073 16:23:54:139
Experiment: 3
Test Point: 18073A3
MFCA1 Gas: 0
MFCA1 Full Scale: 20.000000
MFCA2 Gas: 0
MFCA2 Full Scale: 0.050000
MFCA3 Gas: 0
MFCA3 Full Scale: 0.100000
EField: 1
EFieldPol: 1
Igniter: 1
Burner: 3
Thermocouple: 2
Radiometer: 0
IPSU 1: 1
IPSU 2: 0
IPSU 3: 0
IPSU 4: 0
IPSU 5: 1
IPSU 6: 0
Fiber: 1
PMT 1 Gain: 6.250000
PMT 2 Gain: 6.500000
PMT 3 Gain: 6.500000

Meanwhile, the column headings are found in the second row which are explained in Section 10.2 Data Conversions (non-BRE) of ACME-DOC-010 Rev D, which begins on page 128. The basic conversions of the 125-Hz data to engineering units are provided in this section, but information on temperature conversions are provided in Sections 10.4 and 10.5 which are found in the same document on pages 142-143. Note that E-FIELD Flames did not make use of a thermocouple rake (which were intended for other ACME experiments), so the data in columns TCA, TCB, TCC, and TCD should be ignored. Meanwhile, the coflow burner used in the first round of E-FIELD Flames testing wasn't instrumented with a thermocouple while the gas-jet burner used in the second-round testing did, so the TCBurner column is only relevant for the second-round tests.

As an example, the column headings from 18073A3_data.csv include the following (again displayed in two columns rather than a long row):

Time
MFCA1StaticMixerFlow
MFCA2BypassFlow
MFCA3FuelFlow
IonCurrent
PTA1
PTA2
PTA3
EfieldVoltage
PMT1
PMT2
PMT3
TCA
TCB
TCC
TCD
TC200A
TC200B
TCBurner
Radiometer+60
Radiometer+60CJ
Radiometer+30
Radiometer+30CJ
RadiometerCenter
RadiometerCenterCJ
Radiometer-30
Radiometer-30CJ
Radiometer-60
Radiometer-60CJ
RadiometerOffset
RadiometerOffsetCJ
TCE
Ch20
Ch21
TCCJ
EfieldSetpointOut

Additional conversion is required for the measurements of the gas flow rates found in second through fourth columns of the data file. By column (e.g., in Excel), the flow rates are for:

B	MFCA1StaticMixerFlow	oxidizer (or inert) gas
C	MFCA2BypassFlow	nitrogen
D	MFCA3FuelFlow	fuel

The ACME MFCs are Hastings HFC-D-302 with default standard temperature and pressures (STP) of 0 Celsius and 760 torr for computations of volumetric flow or velocity. It is very important to understand that slpm, i.e., standard liters per minute, is a measurement of the mass (or alternatively molar) flow rate which is independent of the actual temperature or pressure. This is discussed for example in the MFC manual, where chapter 9 (on page 39) is "Volumetric vs. Mass Flow."

Conversions for the actual gases can be made with information provided in the manual on subsequent pages and the spreadsheets GasDataTables.xls and Inside gas mix calculator.xls which were both provided by the MFC manufacturer. First, open GasDataTables.xls and enable content, and then open Inside gas mix calculator.xls and enable content. From this, one should be able to enter mixtures and compute appropriate Gas Conversion Factors (GCFs).

Alternatively, the conversions above can be made with spreadsheets prepared by the ACME engineering team, where options include the following:

ACMEFlowCalculator.xlsx
ACMEFlowAndPressuresRawDataConversion.xlsx

Computation of gas velocities, for example for the gas-jet and coflow burners, depends on burner dimensions. For the coflow burner, ACME mechanical design engineer Adrian Drake indicated the following:

> "The inner diameter of the inner tube is .085 inches. Its cross sectional area is .00567 in^2, or 3.66 x 10^-6 m^2.
>
> The flow of the outer tube is a little tricky to calculate, based on where and how you measure. There's an internal step in the outer tube that can't be seen from outside, and blocks the flow to the outer honeycombs. Because of this, when we calculated the cross-sectional are of the outer tube, it measures from the inner step.
>
> The outer tube's outer diameter at that step is 0.984 inches. Its cross section area at that diameter is .7605 in^2. From this we have to subtract the inner tube's outer diameter: .095 inch diameter, so area is .00709. Subtract from the Outer Diameter and we get .7534 in^2, or .000486 m^2."

The dimensions of the coflow burner's inner tube are documented in greater detail in the ACME Inspection Verification Report (ACME-VER-001 Rev. A, ACME-VER-001_A.pdf) on pages 138-143 of the pdf file. From that section:

> The dimensions specified for this requirement are an I.D. of 0.0827 in. (2.1 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0850 in. (2.159 mm) for the I.D with minimum and maximum of the I.D., 0.0840 in. (2.134 mm) and 0.0860 in. (2.184 mm). The minimum and maximum values fall within the allowable tolerance range of 2.0-2.2 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the ID are met.
>
> The dimensions specified for this requirement are an O.D. of 0.0945 in. (2.4 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0950 in. (2.413 mm) for the O.D with minimum and maximum of the O.D. 0.0940 in. (2.388 mm) & 0.0960 in. (2.438 mm). The minimum and maximum values fall within the allowable tolerance range of 2.3-2.5 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the O.D. are met.

The 2.1-mm gas-jet burner used for the E-FIELD Flames experiment used the same tubing as the coflow burner's inner tube, where the tube dimensions are based on certification by the tubing manufacturer rather than measurement. The tubing dimensions for this burner are documented in greater detail in pages 102-104 of ACME-VER-001_A.pdf. From that section:

> The dimensions specified for this requirement are an I.D. of 0.0827 in. (2.1 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0850 in. (2.159 mm) for the I.D with minimum and maximum of the I.D., 0.0840 in. (2.134 mm) and 0.0860 in. (2.184 mm). The minimum and maximum values fall within the allowable tolerance range of 2.0-2.2 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the ID are met.
>
> The dimensions specified for this requirement are an O.D. of 0.0945 in. (2.4 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0950 in. (2.413 mm) for the O.D with minimum and maximum of the O.D. 0.0940 in. (2.388 mm) and 0.0960 in. (2.438 mm). The minimum and maximum values fall within the allowable tolerance range of 2.3-2.5 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the O.D. are met.

Of course, dimensions are also available from the burner drawings, but the dimensions are often not found on the assembly drawings so the appropriate component drawings must typically be consulted.

2. Mass Flow Controller (MFC) Setpoint File

While the basic data file includes measurement of the gas flow rates, it may be helpful to understand the intended flow rates, where that information is found in a separate file for each test with a name like 18073A3_mfcSetpoints.csv. While it shares strong similarities to the basic data file, the rows of data are not added at a continuous and consistent rate. The file is briefly discussed in section 8.2 Mass Flow Controller Setpoint File on page 124 of ACME-DOC-010 Rev D.

3. Fuel Oxidizer Management Assembly (FOMA) data

Relevant pressure, temperature, etc. data from the CIR is not found in the data output file discussed in section 4.A.1 of this document. Instead, it is found in 1-Hz files with names such as 18073A3_ACME_CAM3_Exposures.txt which are again in a comma-separated values (CSV) format despite the misleading txt suffix. As such, these files can be opened with standard spreadsheet software.

Columns A and B (e.g., in Excel) both include timestamps, which can slightly differ (e.g., by a second), where column A is from the Image Processing and Storage Unit for Gigabit ethernet cameras (IPSU-G, which is also referred to as IPSU-5). The IPSU-G was used to record the image data from the ACME Data Camera (DataCam) and the Cool Flames Investigation (CFI) intensified camera. Meanwhile, column B provides timestamps for the CIR's FOMA Control Unit (FCU), where the FOMA is the Fuel Oxizidizer Management Assembly. The headings for the first two columns are therefore:

IPSU5 TimeStamp	FCU TimeStamp

Columns C-W have data from the IPSUG, of which the most significant is column T, named IPSUG_CAMERA_3_MAN_EXP(N/A). That column provides the ACME DataCam exposure time in microseconds and is the source of the file name. The headings for these columns are listed here in two columns rather than a long row:

IPSUG_SYSTEM_STATE(N/A)	IPSUG_CAMERA_1_GAIN_MODE(N/A)
IPSUG_SYSTEM_5V_POS(Volts)	IPSUG_CAMERA_1_AUTO_GAIN(N/A)
IPSUG_SYSTEM_12V_POS(Volts)	IPSUG_CAMERA_1_MAN_GAIN(N/A)
IPSUG_LV_ACTIVE_CAMERAS(N/A)	IPSUG_CAMERA_3_FRAME_RATE(N/A)
IPSUG_LV_LEFT_CAMERA(N/A)	IPSUG_CAMERA_3_EXP_MODE(N/A)
IPSUG_LV_RIGHT_CAMERA(N/A)	IPSUG_CAMERA_3_AUTO_EXP(N/A)
IPSUG_CAMERA_1_PIXEL_FORMAT(N/A)	IPSUG_CAMERA_3_MAN_EXP(N/A)
IPSUG_CAMERA_1_FRAME_RATE(N/A)	IPSUG_CAMERA_3_GAIN_MODE(N/A)
IPSUG_CAMERA_1_EXP_MODE(N/A)	IPSUG_CAMERA_3_AUTO_GAIN(N/A)
IPSUG_CAMERA_1_AUTO_EXP(N/A)	IPSUG_CAMERA_3_MAN_GAIN(N/A)
IPSUG_CAMERA_1_MAN_EXP(N/A)	

Neglecting the timestamp in column B, the FOMA data begins in column X, where columns X-AR provide measurements from numerous pressure transducers in psia.

Columns AS-BB provide the measurements of the gas flow rates (AS-AW) and their setpoints (AX-BB) in slpm. However, the needed FOMA MFCs were simply forced to be full open when needed, where the flow was more accurately measured and controlled by ACME's MFCs. The column headings are listed below in two columns, where the CIR MFCs are identified per thee ACME/CIR Integrated Pneumatic Schematic, drawing S1101MFS0004_rev_A:

Fuel	FCU_MP_MFC1_RATE(Slpm)	FCU_MP_MFC1_CONTROL(SLPM)
Diluent	FCU_MP_MFC2_RATE(Slpm)	FCU_MP_MFC2_CONTROL(SLPM)
Oxygen	FCU_MP_MFC3_RATE(Slpm)	FCU_MP_MFC3_CONTROL(SLPM)
Nitrogen	FCU_MP_MFC4_RATE(Slpm)	FCU_MP_MFC4_CONTROL(SLPM)
Vent	FCU_MP_MFC5_RATE(Slpm)	FCU_MP_MFC5_CONTROL(SLPM)

It is important to understand that nitrogen was <u>not</u> delivered through MFC4 during ACME tests, but instead through a bypass with no FOMA MFC. ACME had its own MFCs for fuel

(often diluted), oxygen (always diluted with an inert), and nitrogen (from the bypass).

Columns BC-BF provide the combustion chamber temperature in Celsius from four thermistors (abbreviated as TM in the column headings). The column headings are listed below in two columns (instead of a row):

FCU_MP_TM1(Deg C)	FCU_MP_TM3(Deg C)
FCU_MP_TM2(Deg C)	FCU_MP_TM4(Deg C)

Columns BG-FX provide the combustion chamber pressures (in psia) measured by four pressure transducers (abbreviated as PT in the column headings), where there are 30 adjacent columns numbered 0-29 for each of those transducers, namely PT16, PT17, PT28, and PT29. Through a clever approach, the many columns provide these measurements at 30 Hz within a file which is otherwise at 1 Hz. At 1 Hz, this data is also found respectively in columns AJ, AK, AN, and AO. Note that PT28 is an outlier, where it may be appropriate to neglect that transducer's data in computing an average FOMA measurement of the chamber pressure. The chamber pressure is also measured by the ACME hardware, where that measurement is found (in volts) in the basic data previously discussed under the heading PTA3. As an example, the column headings for 30-Hz PT16 data (starting at column BG) are as follows (in three columns):

FCU_PT16_INDEX_0(PSIA)	FCU_PT16_INDEX_10(PSIA)	FCU_PT16_INDEX_20(PSIA)
FCU_PT16_INDEX_1(PSIA)	FCU_PT16_INDEX_11(PSIA)	FCU_PT16_INDEX_21(PSIA)
FCU_PT16_INDEX_2(PSIA)	FCU_PT16_INDEX_12(PSIA)	FCU_PT16_INDEX_22(PSIA)
FCU_PT16_INDEX_3(PSIA)	FCU_PT16_INDEX_13(PSIA)	FCU_PT16_INDEX_23(PSIA)
FCU_PT16_INDEX_4(PSIA)	FCU_PT16_INDEX_14(PSIA)	FCU_PT16_INDEX_24(PSIA)
FCU_PT16_INDEX_5(PSIA)	FCU_PT16_INDEX_15(PSIA)	FCU_PT16_INDEX_25(PSIA)
FCU_PT16_INDEX_6(PSIA)	FCU_PT16_INDEX_16(PSIA)	FCU_PT16_INDEX_26(PSIA)
FCU_PT16_INDEX_7(PSIA)	FCU_PT16_INDEX_17(PSIA)	FCU_PT16_INDEX_27(PSIA)
FCU_PT16_INDEX_8(PSIA)	FCU_PT16_INDEX_18(PSIA)	FCU_PT16_INDEX_28(PSIA)
FCU_PT16_INDEX_9(PSIA)	FCU_PT16_INDEX_19(PSIA)	FCU_PT16_INDEX_29(PSIA)

4. Space Acceleration Measurement System (SAMS) data

The Combustion Integrated Rack (CIR) in which the experiment was conducted, was equipped with a Passive Rack Isolation System (PaRIS), but it was not utilized for ACME testing because of the need for crew time to enable and disable the isolation (by respectively removing and installing alignment guides) on each test day. However, the CIR was equipped with a 3-axis SAMS sensor head. The acceleration archives, along with relevant documentation, are accessible from https://gipoc.grc.nasa.gov/wp/pims/home/, the home page for the Principal Investigator Microgravity Services (PIMS).

The following guidance for the SAMS data was provided by Kenneth Hrovat (NASA Glenn): In addition to the documentation and links from the PIMS home page, we introduce a few ways below to get started with exploring the CIR microgravity environment taking the perspective of SAMS measurements from within the rack.

Roadmap Spectrograms

To get a high-level, spectral roadmap view within two days after measurement, you can:

a. Use your browser and start at this link: http://pims.grc.nasa.gov/roadmap

b. In the upper left pane, you can navigate by GMT date to see a list of plots (PDF files) in the right pane for the GMT date of your choice. We refer to these PDFs as roadmap plots. For SAMS, these are acceleration spectrograms, which for vibratory data are a good way to look at patterns, structures and boundaries in both time and frequency for long time spans -- nominally, 8 hours per plot. These spectrograms are mostly qualitative and crudely quantitative three-dimensional plots in that a color scale attempts to map power spectral density magnitude as a color for the 3rd dimension. *NOTE*: Ticking the Navigation Filter for "es05" in the bottom pane will then only highlight days in yellow in the upper left pane when the operations team powered on the SAMS sensor in the CIR to take measurements.
c. You might want to navigate directly to a GMT day of interest for your science operations or run, but you can also browse day-by-day too in order to get some context comparing various 8-hour spans to get a feel for the environment under various conditions.

Binary Data

To download actual SAMS measurement data for use with your models or to correlate acceleration measurements with your science or support data, you can…

a. Use your browser and start at this link: http://pims.grc.nasa.gov/ftp/pad
b. Navigate by GMT date, for example you can follow year, month and day links to end up at: https://pims.grc.nasa.gov/ftp/pad/year2020/month06/day18/ where you see a list of SAMS sensor directories. The sensor subdirectory of interest for CIR is samses_accel_es05.
c. Once you click into that directory link, you will see data and header files listed for the CIR sensor.
d. The instructions at the following link help you to read these data: http://pims.grc.nasa.gov/plots/user/mohan/how2read_pad_binary_files.html

B. Image Files

Image data and files discussed in this section are associated with the following cameras:

Analog, operations (ops), internal camera
ACME Data Camera (DataCam)
Cool Flames Investigation (CFI) intensified camera
High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package

The Low Light Level Ultraviolet (LLL-UV) camera is not addressed herein because it was not used for E-FIELD Flames or most of the current ACME experiments. Thus far, the Structure and Response of Spherical Diffusion Flames (s-Flame) investigation is the only ACME study to use the LLL-UV camera.

1. Analog, internal, operations (ops) camera

This color analog camera is mounted on the ACME chamber insert, and as such is within the combustion chamber and is sometimes referred to as the internal camera. Its primary purpose is to support the operations, e.g., to visualize the burner and igniter, and as such is also often referred to as the operations or ops camera.

The NTSC video (named for the National Television System Committee) from the ops camera is at 29.97 frames per second (fps) or nominally 30 fps. It is not recorded on the space station but is streamed to the ground in near real time and is recorded on CIR consoles at the Glenn ISS Payload Operations Center (GIPOC) in a Transport Stream (TS) format using VLC Media Player which can be freely downloaded from https://www.videolan.org/vlc/. That software has a snapshot function allowing for convenient extraction of still images, e.g., for making measurements of flame dimensions.

2. ACME Data Camera (DataCam)

ACME's primary camera is a high-definition (1360 x 1024), 12-bit color camera mounted outside the chamber on CIR's optical bench. It is equipped with a motorized lens with independent control of the zoom, iris, and focus. It has a filter barrel with a fused silica window (for color imaging), a blue-green BG7 Schott glass filter, and bandpass filters for imaging CH* and HCHO (formaldehyde) chemiluminescence.

It was characterized prior to launch to enable temperature measurements via soot pyrometry and Thin Filament Pyrometry (TFP), soot volume fraction, and quantitative determination of the CH* concentration.

Soot temperatures can be determined from this camera's images via techniques such as those described in the following references:

a. S. De Iuliis, M. Barbini, S. Benecchi, F. Cignoli, and G. Zizak, Determination of the Soot Volume Fraction in an Ethylene Diffusion Flame by Multiwavelength Analysis of Soot Radiation, Combustion and Flame 115 (1998) 253–261.

b. B. Ma and M. B. Long, Combined Soot Optical Characterization Using 2-D Multi-Angle Light Scattering and Spectrally Resolved Line-of-Sight Attenuation and Its Implication on Soot Color-Ratio Pyrometry, Applied Physics B 117 (2014) 287–303.

c. B. Ma, S. Cao, D. Giassi, D. P. Stocker, F. Takahashi, B. A. V. Bennett, M. D. Smooke, and M. B. Long, An Experimental and Computational Study of Soot Formation in a Coflow Jet Flame under Microgravity and Normal Gravity, Proceedings of the Combustion Institute 35 (2015) 839–846.

d. N. J. Kempema, B. Ma, and M. B. Long, Investigation of In-Flame Soot Optical Properties in Laminar Coflow Diffusion Flames Using Thermophoretic Particle Sampling and Spectral Light Extinction, Applied Physics B 122 (2016).

e. N. J. Kempema and M. B. Long, Effect of Soot Self-Absorption on Color-Ratio Pyrometry in Laminar Coflow Diffusion Flames, Optics Letters 43 (2018) 1103.

Meanwhile, TFP measurements can be made using approaches such as those discussed in:

f. V. Vilimpoc and L. P. Goss, SiC-Based Thin-Filament Pyrometry: Theory and Thermal Properties, Symposium (International) on Combustion 22 (1989) 1907–1914.

g. W. M. Pitts, Thin-Filament Pyrometry in Flickering Laminar Diffusion Flames, Symposium (International) on Combustion 26 (1996) 1171–1179.

h. J. D. Maun, P. B. Sunderland, and D. L. Urban, Thin-Filament Pyrometry with a Digital Still Camera, Applied Optics 46 (2007) 483–488.

i. P. B. Kuhn, B. Ma, B. C. Connelly, M. D. Smooke, and M. B. Long, Soot and Thin-Filament Pyrometry Using a Color Digital Camera, Proceedings of the Combustion Institute 33 (2011) 743–750.

j. B. Ma, G. Wang, G. Magnotti, R. S. Barlow, and M. B. Long, Intensity-Ratio and Color-Ratio Thin-Filament Pyrometry: Uncertainties and Accuracy, Combustion and Flame 161 (2014) 908–916.

Particularly when making quantitative measurements, implementation of an Abel transform or similar is likely appropriate given the axisymmetry of the flames in the E-FIELD Flames investigation. In fact, all six of the current ACME experiments feature axisymmetric or spherically symmetric flames. Approaches are compared in this paper:

k. C. J. Dasch, One-Dimensional Tomography: A Comparison of Abel, Onion-Peeling, and Filtered Backprojection Methods, Applied optics 31 (1992) 1146–1152.

The a-k references above were taken from the following conference paper on early ACME results from the Coflow Laminar Diffusion Flame (CLD Flame) experiment:

Jesse Tinajero, Davide Giassi, Dennis Stocker, and Marshall Long. “Experimental Study on the Influence of Gravity on Highly Diluted and Sooting Coflow Flames,” 11th US National Combustion Meeting of the Combustion Institute, Pasadena, CA, 24-27 March 2019.

Meanwhile, this paper on an ISS precursor to the CLD Flame experiment is an appropriate reference for determination of CH* concentration.

D. Giassi, S. Cao, B.A. Bennett, D.P. Stocker, F. Takahashi, M.D. Smooke, and M.B. Long, “Analysis of CH* Concentration and Flame Heat Release Rate in Laminar Coflow Diffusion Flames under Microgravity and Normal Gravity,” *Combustion and Flame*, Vol. 167, pp. 198-206 (2016).

Images are normally downlinked within a week of the testing, but selected images are often downlinked during a test day. Regardless, the images are downloaded in a wrapped Halcon object (HOBJ) format, which are not readily opened with typical image processing software.

Most ACME investigators are processing the HOBJ files using the OMA2 software developed by Prof. Marshall Long (Yale U.), the lead for ACME’s CLD Flame experiment. The OMA2 software is freely available in both Mac and cross-platform (Qt) versions, where the latter can be used with both PC and Linux operating systems.

The OMA2 read-hobj function was also rewritten as a Python function by Dr. Jesse Tinajero (Yale U.), where Python is an open source programming language that is easily obtainable for MacOS, Linux, and Windows operating systems. The Python script to open the HOBJ files can be downloaded from https://github.com/mblong/hobjExample.

The HOBJ files can also be readily viewed and processed using ImageJ or its preloaded Fiji variant, where both can be freely downloaded. In either software package, the HOBJ files can imported (rather than opened) as raw files with the following settings:

Image type	16-bit unsigned
Width	1360
Height	1024
Offset to first image	6244

However, imported DataCam images will be monochrome unless a Bayer filter (e.g., as described at https://en.wikipedia.org/wiki/Bayer_filter) is subsequently implemented. It should be possible to install a plug-in to accomplish that color conversion. As another option, Python and Matlab both have image demosaicing functions.

NASA is also working (at the time of this writing) toward providing converted images from the ACME DataCam, where the general priority is to support the ongoing ACME testing rather than covert images for experiments like E-FIELD Flames which have already been completed.

In order to use the ACME DataCam for quantitative measurements it is critical to avoid saturation, for example by using a camera gain of zero. Another employed approach was to not use the camera's white balance which by default multiplies the red and blue by 1.41 and 1.68, respectively, because that could inadvertently lead to saturation. Such multiplication can be done after the HOBJ file is converted and the colors demosaiced, where in OMA2 the white balancing is accomplished with the commands:

```
*rgb 1.41 1 1.68
clip 4095
```

In summary, the white balance, if desired, needs to be done after the fact through software because it was generally not done with the hardware to avoid the risk of saturation.

Note that the ACME images will be naturally dark because of the use of the 16-bit file format with 12-bit image data. As such, adjustments to the image brightness will generally be necessary even if the images are not demosaiced. Furthermore, cameras' CCD arrays are susceptible to radiation damage while in orbit. As a result, the images feature hot pixels which increase in number over the months and years in space. As such, it will appear as if there is a star field within the combustion chamber. Therefore, the images should ideally be processed to remove the bright outlying pixels. The radiation damage is also obvious in the ops camera imaging.

Although the DataCam was typically set for imaging at 30 fps, that is trumped by the exposure settings which were often long because of the dim nature of the microgravity flames. As a result, the framing rate was typically far less. Different exposures could have been selected for different phases of the tests, e.g., prior to ignition and during the test, so the framing rate may not be consistent. Bracketed imaging could also be employed, where a repeating sequence of different exposure durations is used to enable appropriate photography under uncertain or varying flame intensities. Bracketing is also used to simultaneously accomplish multiple imaging objectives, e.g., to appropriately image both brightly glowing soot (or TFP fibers) as well as the much dimmer blue portion of the flame.

In interpreting the image data, it may be helpful to check the ACME DataCam metadata file

which should be available for each successful flame test, e.g., to verify the exposure duration. The formatting of this file changed during early 2018 (between mid-March and early April), when the first round of E-FIELD Flames tests (using the coflow burner) was being conducted on the ISS. Therefore, examples of early and later versions are presented, where their suffixes were respectively metadata and mdipsug. However, both are in a comma-delimited (CSV) format and can be opened with spreadsheet software or alternately as a text file. Column headings are found in the first row of such files.

Early version of the ACME DataCam metadata

An early example is 20180314_162445.789_ACME_18073A3.metadata where ACME designates the DataCam. Presented below are the column headings and example values from the test's first image (numbered 000000), and the corresponding camera parameter and a description provided by ACME software engineer Jeffrey Eggers:

Filename /sd0/sci/18073A3/20180314_162445.925_ACME_18073A3_00000.hobj
Camera parameter: n/a
Image path and file name with timestamp of when the image was grabbed from the camera.

BufferIncompleteTotal 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received.

BufferIncomplete 0
Camera parameter: GtlBufferIncomplete
State of the current buffer. In case of an incomplete buffer 1 is returned. When using the filter driver incomplete buffers are silently discarded. Check the GtlBlockID to see if buffers are missing.

BuffersNum 5
Camera parameter: GtlNumBuffers
Number of currently used buffers for the internal ring buffer.

BlockId 690
Camera parameter: GtlBlockID
ID of the last received block (typically of type image) as reported by the device. If no block has been received, 0 is returned.

TickTimestamp 14178831461500
Camera parameter: GtlBufferTimestamp
Time stamp of the current buffer in ticks. The time in seconds can be calculated via the parameter 'GevTimestampTickFrequency.'

Timestamp 20180314_162445.925
Camera parameter: n/a
The timestamp portion of the filename, when the image was grabbed from the camera.

Milliseconds 136
Camera parameter: n/a
Difference between last image timestamp and current image timestamp.

CalculatedRate 0
Camera parameter: n/a
This is GevTimestampTickFrequency / (GtlBufferTimestamp - GtlBufferTimestamp of last image). GevTimestampTickFrequency is querried each image.

BinningX 1
Camera parameter: BinningHorizontal
Binning in horizontal direction. Binning is the summing of charge of adjacent pixels.
BinningY 1
Camera parameter: BinningVertical
Binning in vertical direction. Binning is the summing of charge of adjacent pixels.
PixelFormat BayerRG12
Camera parameter: PixelFormat
Format of the image data.
Exposure 100000
Camera parameter: ExposureTimeAbs
Exposure duration, in microseconds.
FrameRate 30.0003
Camera parameter: AcquisitionFrameRateAbs
Frame rate, in frames per second. This is applicable when either the FrameStart trigger mode is disabled, or the FrameStart trigger source is FixedRate. Depending on the exposure duration, the camera may not achieve the frame rate set here.
Gain 0
Camera parameter: GainRaw
Gain value of analog A/D stage. Units are in dB.
Height 1024
Camera parameter: Height
Height of image, in pixels.
Width 1360
Camera parameter: Width
Width of image, in pixels.
RegionX 0
Camera parameter: OffsetX
Starting column of the readout region (relative to the first column of the sensor) in pixels.
RegionY 0
Camera parameter: OffsetY
Starting row of the readout region (relative to the first row of the sensor) in pixels.

Later version of the ACME DataCam metadata

This is an example of the later metadata format from ACME_18269A1.mdipsug. Little has changed from the earlier format other than the order of the parameters. Again, the column headings are presented with the values for the first image (i.e., from the file's second row).

Timestamp 20180926_152820.593
Filename /sd1/sci/18269A1/20180926_152820.593_ACME_18269A1_00000.hobj
DeltaMilliseconds 236
CalculatedRate 0
BinningX 1
BinningY 1
BlockId 2
BufferIncomplete 0
BufferIncompleteTotal 0

BuffersNum	5
ExposureManual	100000
FrameRate	30.0003
GainManual	0
Height	1024
PixelFormat	BayerRG12
RegionX	0
RegionY	0
TickTimestamp	5112531190000
Width	1360

Regardless of the version, it is very important to understand that the exposure time, in microseconds, trumps the specified framing rate (in fps), i.e., where long exposures will lead to a shorter actual framing rate. In this example case, the 0.1-s exposure time means that the framing rate (at the test's start) was nominally 10 fps rather than the 30 indicated by FrameRate. It is furthermore important to understand that the exposure time (and thus actual framing rate) was often varied during the test, where that could be seldom or often if bracketing was employed, i.e., where a repeated sequence of images was taken at different exposures.

For the DataCam, the exposures times, etc. can also be found in files with names like 18073A3_ACME_CAM3_Exposures.txt, where such files are discussed in section 4.A.3 of this document because (despite the file naming) they primarily include pressures, flow rates, and temperatures from the CIR's Fuel Oxidizer Management Assembly (FOMA).

The parameter files, with names such as 18073A3_VoltageSweep.txt, control the DataCam exposures and much more. However, these files were not always saved as intended and were sometimes corrected after the fact. In contrast, the text-file logs, with names such as 2018-10-17_12-26ACME.log, are a reliable source for the parameters which specified the imaging settings and much more. There is a single file of this type for each test day (at least normally) and the parameters are a small fraction of the automated log content, but one can search for a test by its number. Log files like this are discussed in section 4.C.4 of this document.

Due to the use of a turning mirror, the DataCam images should be flipped vertically if upright images are desired, i.e., where the burner is at the bottom and the flame extends upward. In this regard, note that the DataCam provides images in a 'landscape' format with the image's long dimension orthogonal to the burner and chamber axis. This is contrary to an ACME science requirement specifying that the long dimension be aligned with the experimental axis to provide images in a 'portrait' format. In this regard, the science requirements were not always met, e.g., because of design tradeoffs, etc.

There is some variation in the ACME camera zoom, where its setting is not completely reproducible. Therefore, the scaling of the image data should be established for each day of testing, or each subset of tests if more than one zoom setting was used within a day. The scaling can typically be carried out using images regularly taken of the ceramic fibers for Thin Filament Pyrometry (TFP), where such imaging is normally made at two fiber array positions. However, the fiber array was not used for the E-FIELD Flames experiment because of the

interference of the electric field and electrode mesh. Therefore, scaling must be carried out using the burner or other visible hardware. Meanwhile, the other cameras had fixed lenses.

Note that files named for example 20180314_162445.789_ACME_18073A3.next are not useful for data interpretation, where such text files simply indicate the next ACME image number that would have been archived, i.e., until that test's image archiving came to an end.

3. Cool Flames Investigation (CFI) intensified camera

This intensified camera was typically configured with a CH* bandpass filter and was generally archived at 23 fps. The grayscale image files are in a Tagged Image File Format (TIFF) format and can be opened readily, where brightness/contrast adjustment may again be required.

While it wasn't an issue for E-FIELD Flames, this camera's Field of View (FOV) is partially blocked by the arms of the fiber array for Thin Filament Pyrometry (TFP) when it is moved to position the fibers across the flame. But E-FIELD Flames didn't use the fiber array because of interference with the electrode mesh and the electric field itself.

As with the ACME DataCam, it might be useful to check the metadata for the CFI intensified camera. This was similarly revised in early 2018 (between mid-March and early April) when the first round of E-FIELD Flames tests (with the coflow burner) was being conducted, such that there are two version of this file relevant to the experiment. Although they are largely similar, the latter has two additional parameters, BitShift and CameraSpeed, and some additional changes, e.g., where some parameter names changed and where the parameter order has changed. Therefore, detailed information on both versions will be presented where the parameter descriptions and names of the corresponding camera parameters were provided by ACME software engineer Jeffrey Eggers. Both versions are in a comma-delimited (CSV) format and can be opened with spreadsheet software or alternately as a text file.

<u>Early version of the CFI intensified camera metadata</u>

An example file name is 20180314_162445.789_INTNS_18073A3.metadata where INTNS designates the CFI camera. Presented below are the column headings and example values from the test's first image (numbered 000000), and the corresponding camera parameter name and a description:

Filename /sd0/sci/18073A3/20180314_162445.861_INTNS_18073A3_00000.tiff
Camera parameter: n/a
Image path and file name with timestamp of when the image was grabbed from the camera.

BufferIncompleteTotal 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received.

BufferIncomplete 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received (where the Camera Parameter should be GtlBufferIncomplete but is GtlIncompleteBuffers instead).

BuffersNum 5
Camera parameter: GtlNumBuffers
Number of currently used buffers for the internal ring buffer.

BlockId 1

Camera parameter: GtlBlockID

ID of the last received block (typically of type image) as reported by the device. If no block has been received, 0 is returned.

TickTimestamp 28342240286

Camera parameter: GtlBufferTimestamp

Time stamp of the current buffer in ticks. The time in seconds can be calculated via the parameter 'GevTimestampTickFrequency.'

Timestamp 20180314_162445.861

Camera parameter: n/a

The timestamp portion of the filename, when the image was grabbed from the camera.

Milliseconds 72

Camera parameter: n/a

Difference between last image timestamp and current image timestamp.

CalculatedRate 0

Camera parameter: n/a

This is GevTimestampTickFrequency / (GtlBufferTimestamp - GtlBufferTimestamp of last image). GevTimestampTickFrequency is querried each image.

BinningX x1

Camera parameter: BinningHorizontal

The number of horizontal photo-sensitive cells that must be combined (added) together. Note that the user has to set the Width feature accordingly.

BinningY x1

Camera parameter: BinningVertical

The number of vertical photo-sensitive cells that must be combined (added) together. Note that the user has to set the Height feature accordingly.

PixelFormat Mono16

Camera parameter: PixelFormat

The format of the pixel to use during the acquisition.

ExposureMode 0

Camera parameter: AecEnable

Whether or not the Automatic Exposure Control function is enabled.

Exposure 32000

Camera parameter: ExposureTimeRaw

The Exposure time in units of uSeconds when ExposureMode is Timed (and AEC is disabled).

CurrentAutoExposure 43067

Camera parameter: CurrentAecExposure

The current AEC exposure value.

FrameRate 23

Camera parameter: CurrentFrameRate

The current frame rate in integer units of frames per second. This is a dynamic value that changes as a function of the camera's mode of operation.

GainMode 0

Camera parameter: AgcEnable

Whether or not the Automatic Gain Control function is enabled.

Gain 143
Camera parameter: GainRaw
The selected analog gain as a raw integer value (when AGC is disabled).
CurrentAutoGain 143
Camera parameter: CurrentAgcGain
The current AGC analog gain value.
RegionHeight 1040
Camera parameter: Height
The actual image height expelled by the camera of the master AOI (in pixels).
RegionWidth 1392
Camera parameter: Width
The actual image width expelled by the camera of the master AOI (in pixels).
RegionX 0
Camera parameter: OffsetX
The horizontal offset from the origin to the master AOI (in pixels).
RegionY 0
Camera parameter: OffsetY
The vertical offset from the origin to the master AOI (in pixels).
ImageHeight 1040
Camera parameter: SensorHeight
The effective height of the sensor in pixels.
ImageWidth 1392
Camera parameter: SensorWidth
The effective width of the sensor in pixels.

Later version of the CFI intensified camera metadata

An example file name is INTNS_18269A1.mdipsug where INTNS designates the CFI intensified camera. Presented below are the column headings and example values from the test's first image (numbered 000000), and the corresponding camera parameter name and a description:

Timestamp 20180926_152820.469
Camera parameter: n/a
The timestamp portion of the filename, when the image was grabbed from the camera.
Filename /sd1/sci/18269A1/20180926_152820.469_INTNS_18269A1_00000.tiff
Camera parameter:
Image path and file name with timestamp of when the image was grabbed from the camera
DeltaMilliseconds 112
Camera parameter: n/a
Difference between last image timestamp and current image timestamp.
CalculatedRate 0
Camera parameter: n/a
This is GevTimestampTickFrequency / (GtlBufferTimestamp - GtlBufferTimestamp of last image). GevTimestampTickFrequency is querried each image.

BinningX x1
Camera parameter: BinningHorizontal
The number of horizontal photo-sensitive cells that must be combined (added) together. Note that the user has to set the Width feature accordingly.

BinningY x1
Camera parameter: BinningVertical
The number of vertical photo-sensitive cells that must be combined (added) together. Note that the user has to set the Height feature accordingly.

BitShift NoShift
Camera parameter: BitShift
The number of bits to shift in the data output.

BlockId 2
Camera parameter: GtlBlockID
ID of the last received block (typically of type image) as reported by the device. If no block has been received, 0 is returned.

BufferIncomplete 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received (where the Camera Parameter should be GtlBufferIncomplete but is GtlIncompleteBuffers instead).

BufferIncompleteTotal 0
Camera parameter: GtlIncompleteBuffers
Current number of incomplete buffers received.

BuffersNum 5
Camera parameter: GtlNumBuffers
Number of currently used buffers for the internal ring buffer.

CameraSpeed Normal
Camera parameter: CameraSpeed
The camera speed.

ExposureAuto 43067
Camera parameter: CurrentAecExposure
The current AEC exposure value.

ExposureManual 32000
Camera parameter: ExposureTimeRaw
The Exposure time in units of uSeconds when ExposureMode is Timed (and AEC is disabled).

ExposureMode 0
Camera parameter: AecEnable
Whether or not the Automatic Exposure Control function is enabled.

FrameRate 23
Camera parameter: CurrentFrameRate
The current frame rate in integer units of frames per second. This is a dynamic value that changes as a function of the camera's mode of operation.

GainAuto 143
Camera parameter: CurrentAgcGain
The current AGC analog gain value.

GainManual 143
Camera parameter: GainRaw
The selected analog gain as a raw integer value (when AGC is disabled).
GainMode 0
Camera parameter: AgcEnable
Whether or not the Automatic Gain Control function is enabled.
ImageHeight 1040
Camera parameter: SensorHeight
The effective height of the sensor in pixels.
ImageWidth 1392
Camera parameter: SensorWidth
The effective width of the sensor in pixels.
PixelFormat Mono16
Camera parameter: PixelFormat
The format of the pixel to use during the acquisition.
RegionHeight 1040
Camera parameter: Height
The actual image height expelled by the camera of the master AOI (in pixels).
RegionWidth 1392
Camera parameter: Width
The actual image width expelled by the camera of the master AOI (in pixels).
RegionX 0
Camera parameter: OffsetX
The horizontal offset from the origin to the master AOI (in pixels).
RegionY 0
Camera parameter: OffsetY
The vertical offset from the origin to the master AOI (in pixels).
TickTimestamp 10391712828
Camera parameter: GtlBufferTimestamp
Time stamp of the current buffer in ticks. The time in seconds can be calculated via the parameter 'GevTimestampTickFrequency'.

Note that 'next' files named for example 20180314_162445.789_INTNS_18073A3.next are not useful for data interpretation, where such text files simply indicate the next image number that would have been archived, i.e., until that test's image archiving came to an end.

4. High Bit-depth Multi-spectral (HiBMs) camera and CIR illumination package

The HiBMs camera was used in combination with an opposed backlight, i.e., the CIR's collimated illumination package. The resultant imaging was used in preceding droplet combustion experiments for measuring droplet diameter, but it may also be useful for determining the soot volume fraction via light extinction. However, the camera's red filter, corresponding to the red illumination, is broad enough to permit significant soot incandescence to pass through the filter. As a result, this imaging is unlikely to be useful for soot volume fraction measurements for all conditions. After post-processing on the ground, the images were made available in the TIFF format and as such are readily opened. The grayscale images were typically archived at 15 fps.

It is important to understand that this camera's images are notably of limited value for E-FIELD Flames because the bracket for the electrode mesh significantly blocked the illumination. ACME's cameras and other optical instruments encircle the burner and made it difficult to avoid optical interference in all circumstances, where design trade-offs were necessary. However, the soot volume fraction can also be computed using images from the ACME DataCam as discussed in that camera's section.

In initial ACME testing with the Coflow Laminar Diffusion Flame (CLD Flame) experiment, the monochrome camera was used without a filter, but that was abandoned because the camera is nearly insensitive to blue. The red filter was used throughout E-FIELD Flames testing.

C. Test Parameters and Other Files

Files discussed in this section include the following:

Test parameter file
Notes
Test timing
Logs

1. Test parameter file

Each E-FIELD Flames test was controlled by a software script and corresponding parameter file, named for example 18073A3_VoltageSweep.txt, specifying settings for imaging, ignition, gas flow, and more. The parameter file and script are discussed extensively in the ACME/E-FIELD Flames Experimental Procedure document, which should be referred to for information on this file type. But note that the ACME.log files should be the best way to verify the as-run test parameters, where the separate parameter files are unfortunately not completely reliable. They were not always saved and as a result were sometimes recreated after the fact.

2. Notes

Notes of various types are available for test days, where not all types were collected or necessarily prepared for each test day. It is important to understand that the notes, whether handwritten or typed, may include mistakes. It is better to rely on automated logs, metadata files, and parameter files or of course the measurements themselves when possible.

The daily test point logs, with names such as '18087 daily test point log.pdf,' chronologically list the tests and the image data acquired which was to be downlinked. Typically, image data was not downlinked without successful, i.e., sustained, ignition. As an example, image data wasn't downlinked if the test aborted prior to ignition. However, the reverse is not always true. Image data was sometimes downlinked when ignition was unsuccessful and as a result the test point log is only a clue as to what transpired. Furthermore, note that image data from certain selected cameras, typically the HiBMs and/or CFI intensified camera(s), was sometimes not downlinked due to the limit of how much data could be reasonably downlinked in a subsequent two-shift day.

Notes with names such as '18087 efield & velocities traces.pdf' include preliminary plots of the electrode mesh voltage and flow rates (as seen on a console during testing), typically along with additional information about the tests.

The test parameter notes, with names such as '18087 test parameter notes.pdf,' show the parameter values which were planned in advance of the test day and the changes made on the test day.

The test sheets, with names such as '18087 test sheets.pdf,' are summary notes about the conduct of each tests conducted that day.

3. Test timing

Timing files with names such as 18073A3_TP.txt can be a convenient way to determine the phases of the test in which the images were acquired. An example E-FIELD Flames file is presented here:

Test Point: 18073A3

Script start: 2018-03-14 16:23:54
Script paused #1: 2018-03-14 16:24:03
Script resumed #1: 2018-03-14 16:24:19
Pre-ignition data: 2018-03-14 16:24:36
Pre-ignition images: 2018-03-14 16:24:46
Preheat igniter: ***
MFCs on for priming: 2018-03-14 16:24:58
Ignition and retraction: 2018-03-14 16:25:03
Flame stabilization #1: ***
Dilution of fuel and inert: 2018-03-14 16:25:13
Flame stabilization #2: 2018-03-14 16:25:13
Ramp coflow to match test cond: 2018-03-14 16:25:13
Flame stabilization #3: 2018-03-14 16:25:13
Ramp to final velocity: ***
Constant flow: ***
MFCs off: 2018-03-14 16:26:23
Post-flame images: 2018-03-14 16:26:23
Post-flame data: 2018-03-14 16:26:26
Script complete: 2018-03-14 16:28:16
Extinction detected: ***
Script paused #2: 2018-03-14 16:26:32
Script resumed #2: 2018-03-14 16:27:13

Camera	Archived	Start Image	End Image
IPSU1	1506	737	1,506
IPSU2	***	***	***
ACME	319	291	319
Intensified	2326	1,143	2,326

4. Logs

There are four primary types of automated logs of which two, with names like the following, may be helpful.

2018-10-17_12-26ACME.log
2018-10-17_12-51_IPSUG.log

As a reminder, the IPSU-G is the processing and storage unit for the images from the ACME DataCam and CFI intensified camera.

Meanwhile, the two logs of commands, with names like the following, are unlikely to be useful for interpreting the E-FIELD Flames results.

2018-10-17_12-59ACMECommands.log
2018-10-17_12-51_IPSUGCommands.log

Each of the four log files are created for each day of testing, rather than for each test. They are text files (despite the log suffix) and as such can be readily opened.

The ACME.log files should be the best way to verify the test parameters, where the separate as-run parameter files are unfortunately not completely reliable. A portion of the log corresponding to a single test, 18290F1, is presented here, where this can best be interpreted after becoming familiar with the contents of the ACME/E-FIELD Flames Experimental Procedure document. But the test parameters and timing of the test are obvious in the excerpt.

<u>ACME.log excerpt</u>

2018-10-17 290 17:41:58:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:41:58:813~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"testpointName":"18290F1"})"
2018-10-17 290 17:42:00:012~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:00:012~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"useBracketedExposures":true})"
2018-10-17 290 17:42:00:912~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:00:912~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"bracketingNumberOfExposures":6})"
2018-10-17 290 17:42:01:912~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:01:912~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"bracketingExposureStartUsec":75})"
2018-10-17 290 17:42:02:912~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:02:912~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"bracketingExposureIncrementUsec":340})"
2018-10-17 290 17:42:03:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:03:813~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"igniterPreheatPowerV":0.0})"
2018-10-17 290 17:42:04:712~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:04:712~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"pmt1Gain":4.75})"
2018-10-17 290 17:42:05:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:05:812~acme~"Received command:

ChangeScriptParameter(parameterInfo:{"pmt2Gain":3.25})"
2018-10-17 290 17:42:06:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:06:812~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"pmt3Gain":3.50})"
2018-10-17 290 17:42:07:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:07:812~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"velocityMixIgnitionMPerSec":0.19})"
2018-10-17 290 17:42:08:712~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:08:712~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"velocityMixFinalMPerSec":0.19})"
2018-10-17 290 17:42:09:712~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:09:712~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldCycles":1})"
2018-10-17 290 17:42:10:712~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:10:712~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldStepVoltage":-350})"
2018-10-17 290 17:42:12:412~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:12:412~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldStepDurationSecs":2})"
2018-10-17 290 17:42:13:112~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:13:112~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldRampRatePerSec":0})"
2018-10-17 290 17:42:13:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:13:812~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldRampDurationSecs":0})"
2018-10-17 290 17:42:14:812~zt.script~*****handleScriptParameterChange
2018-10-17 290 17:42:14:812~acme~"Received command:
ChangeScriptParameter(parameterInfo:{"efieldHoldDurationSecs":0})"
2018-10-17 290 17:42:43:311~acme~"Received command: SystemRunScript()"
2018-10-17 290 17:42:43:312~zt.script~param: "testpointName":"18290F1"
2018-10-17 290 17:42:43:312~zt.script~param: "preIgnitionDataTimeSec":0
2018-10-17 290 17:42:43:312~zt.script~param: "postFlameDataTimeSec":5
2018-10-17 290 17:42:43:312~zt.script~param: "useIPSUs":true
2018-10-17 290 17:42:43:312~zt.script~param: "ipsuTimeoutSec":40
2018-10-17 290 17:42:43:312~zt.script~param: "useAcmeCamera":true
2018-10-17 290 17:42:43:312~zt.script~param: "useIntensifiedCamera":true
2018-10-17 290 17:42:43:312~zt.script~param: "ipsu1ArchiveRate":2
2018-10-17 290 17:42:43:312~zt.script~param:
"dataCameraPreIgnitionExposureUs":100000
2018-10-17 290 17:42:43:312~zt.script~param: "dataCameraTestExposureUs":20000
2018-10-17 290 17:42:43:312~zt.script~param: "analogCameraPreIgnitionShutter":2
2018-10-17 290 17:42:43:312~zt.script~param: "analogCameraTestShutter":5
2018-10-17 290 17:42:43:312~zt.script~param: "intensifierGainAmplitude":25000
2018-10-17 290 17:42:43:312~zt.script~param: "preIgnitionImageTimeSec":0
2018-10-17 290 17:42:43:312~zt.script~param: "postFlameImagingTimeSec":3
2018-10-17 290 17:42:43:313~zt.script~param: "useSideLedBank":true
2018-10-17 290 17:42:43:313~zt.script~param: "useEndLedBank":true

2018-10-17 290 17:42:43:313~zt.script~param: "useBracketedExposures":true
2018-10-17 290 17:42:43:313~zt.script~param: "bracketingImagesPerExposure":1
2018-10-17 290 17:42:43:313~zt.script~param: "bracketingNumberOfExposures":6
2018-10-17 290 17:42:43:313~zt.script~param: "bracketingExposureStartUsec":75
2018-10-17 290 17:42:43:313~zt.script~param: "bracketingExposureIncrementUsec":340
2018-10-17 290 17:42:43:313~zt.script~param: "igniterIgnitionPosition":2472
2018-10-17 290 17:42:43:313~zt.script~param: "igniterPreheatPowerV":0
2018-10-17 290 17:42:43:313~zt.script~param: "igniterPreheatTimeMs":0
2018-10-17 290 17:42:43:313~zt.script~param: "igniterIgnitionPowerV":3.75
2018-10-17 290 17:42:43:313~zt.script~param: "igniterIgnitionDurationMs":2000
2018-10-17 290 17:42:43:313~zt.script~param: "igniterDelayBeforeRetractMs":1500
2018-10-17 290 17:42:43:313~zt.script~param: "useIgnitionDetection":true
2018-10-17 290 17:42:43:313~zt.script~param: "extinctionDetectionStopsTest":true
2018-10-17 290 17:42:43:313~zt.script~param: "pmt1Gain":4.75
2018-10-17 290 17:42:43:313~zt.script~param: "pmt2Gain":3.25
2018-10-17 290 17:42:43:313~zt.script~param: "pmt3Gain":3.5
2018-10-17 290 17:42:43:314~zt.script~param: "useFoma":true
2018-10-17 290 17:42:43:314~zt.script~param: "flowTimeoutSec":150
2018-10-17 290 17:42:43:314~zt.script~param: "nitrogenPrimingDurationSec":10
2018-10-17 290 17:42:43:314~zt.script~param: "primingAndPreIgnitionDurationSec":5
2018-10-17 290 17:42:43:314~zt.script~param: "stabilizationTime1Sec":1.5
2018-10-17 290 17:42:43:314~zt.script~param: "dilutionTimeSec":0
2018-10-17 290 17:42:43:314~zt.script~param: "stabilizationTime2Sec":0
2018-10-17 290 17:42:43:314~zt.script~param: "matchingCoflowRampTimeSec":0
2018-10-17 290 17:42:43:314~zt.script~param: "stabilizationTime3Sec":0
2018-10-17 290 17:42:43:314~zt.script~param: "fuelAndCoflowRampTimeSec":0
2018-10-17 290 17:42:43:314~zt.script~param: "constantFlowTimeSec":5
2018-10-17 290 17:42:43:314~zt.script~param: "velocityAirIgnitionMPerSec":0
2018-10-17 290 17:42:43:314~zt.script~param: "velocityAirFinalMPerSec":0
2018-10-17 290 17:42:43:314~zt.script~param: "velocityMixIgnitionMPerSec":0.19
2018-10-17 290 17:42:43:314~zt.script~param: "velocityMixFinalMPerSec":0.19
2018-10-17 290 17:42:43:314~zt.script~param: "concentrationInitial":0.8
2018-10-17 290 17:42:43:314~zt.script~param: "concentrationFinal":0.8
2018-10-17 290 17:42:43:314~zt.script~param: "rampCount":0
2018-10-17 290 17:42:43:315~zt.script~param: "rampSizeMPerSec":0.05
2018-10-17 290 17:42:43:315~zt.script~param: "rampTimeSec":10
2018-10-17 290 17:42:43:315~zt.script~param: "rampConstantFlowTimeSec":10
2018-10-17 290 17:42:43:315~zt.script~param: "efieldCommandManually":false
2018-10-17 290 17:42:43:315~zt.script~param: "efieldInitializeVolts":-800
2018-10-17 290 17:42:43:315~zt.script~param: "efieldInitializeTimeSec":2
2018-10-17 290 17:42:43:315~zt.script~param: "efieldMinVoltage":-10000
2018-10-17 290 17:42:43:315~zt.script~param: "efieldMaxVoltage":-1000
2018-10-17 290 17:42:43:315~zt.script~param: "efieldCycles":1
2018-10-17 290 17:42:43:315~zt.script~param: "efieldStepVoltage":-350
2018-10-17 290 17:42:43:315~zt.script~param: "efieldStepDurationSecs":2
2018-10-17 290 17:42:43:315~zt.script~param: "efieldRampRatePerSec":0
2018-10-17 290 17:42:43:315~zt.script~param: "efieldRampDurationSecs":0

2018-10-17 290 17:42:43:315~zt.script~param: "efieldHoldDurationSecs":0
2018-10-17 290 17:42:43:315~zt.script~E-Field Voltage Sweep test point 18290F1 STARTED
2018-10-17 290 17:42:43:315~zt.script~Setting PMT gains
2018-10-17 290 17:42:43:316~zt.script~priming/ignition flow mfca1/coflow = 0 N2 SLPM
2018-10-17 290 17:42:43:316~acme.diag~"Received command: PmtSetAllGains(PMT1Gain:4.75,PMT2Gain:3.25,PMT3Gain:3.5)"
2018-10-17 290 17:42:43:316~zt.script~priming/ignition flow mfca2/inert = 0.007234852958624181 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~priming/ignition flow mfca3/fuel = 0.047739049545524145 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~matched flow mfca1/coflow = 0 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~diluted igntion flow mfca2/inert = 0.007234852958624181 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~diluted ignition flow mfca3/fuel = 0.047739049545524145 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~coflow acceration = 0 m/s2
2018-10-17 290 17:42:43:316~zt.script~final flow mfca1/coflow = 0 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~final flow mfca2/inert = 0.007234852958624181 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~final flow mfca3/fuel = 0.047739049545524145 N2 SLPM
2018-10-17 290 17:42:43:316~acme~"Received command: SystemOpenTestPoint(testpointName:18290F1)"
2018-10-17 290 17:42:43:316~zt.script~ramp flow mfca1/coflow = 0 N2 SLPM
2018-10-17 290 17:42:43:316~zt.script~ramp flow mfca2/inert = 0.0019039086733221532 N2 SLPM
2018-10-17 290 17:42:43:317~zt.script~ramp flow mfca3/fuel = 0.012562907775137933 N2 SLPM
2018-10-17 290 17:42:43:317~zt.script~Changing cameras to pre-ignition settings
2018-10-17 290 17:42:43:317~zt.script~Intensified camera to safe mode
2018-10-17 290 17:42:43:317~acme.image~"Received command: DataCameraSetManualExposure(integrationTimeUs:100000)"
2018-10-17 290 17:42:43:318~acme.image~"Received command: AnalogCameraSetShutter(shutter:2)"
2018-10-17 290 17:42:43:353~acme.image~Analog command successful.
2018-10-17 290 17:42:43:353~acme.image~"Received command: IntensifiedCameraGate(gateType:0,pulsedGateWidthCounts:0)"
2018-10-17 290 17:42:43:354~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:42:43:354~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:42:43:427~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:42:43:429~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:42:43:554~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:42:43:793~zt.fcf~Success sending RVC command in rvcThreadWaiting on RVC response in rvcThread
2018-10-17 290 17:42:43:794~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:42:43:817~acme.image~"Received command: IntensifiedCameraGainAmplitude(gainAmplitudeCounts:2048)"

2018-10-17 290 17:42:43:818~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:42:43:827~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:42:43:829~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:42:45:817~zt.script~Inserting igniter
2018-10-17 290 17:42:45:817~acme.image~"Received command:
IlluminationConfigure(sideLEDBankState:1,endLEDBankState:1)"
2018-10-17 290 17:42:45:818~zt.script~Clearing MFC integrated flow
2018-10-17 290 17:42:45:818~acme.igniter~"Received command:
IgniterMove(position:2472)"
2018-10-17 290 17:42:46:070~acme.diag~"Received command:
ExtinctionDetectionSetState(state:0)"
2018-10-17 290 17:42:46:071~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*00g31=0)"
2018-10-17 290 17:42:46:120~acme.flow~Reply for command: *00g31=0
2018-10-17 290 17:42:46:120~acme.flow~">"
2018-10-17 290 17:42:46:818~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*01g31=0)"
2018-10-17 290 17:42:47:031~acme.flow~Reply for command: *01g31=0
2018-10-17 290 17:42:47:031~acme.flow~">"
2018-10-17 290 17:42:47:818~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31=0)"
2018-10-17 290 17:42:48:008~acme.flow~Reply for command: *02g31=0
2018-10-17 290 17:42:48:008~acme.flow~">"
2018-10-17 290 17:42:48:818~zt.script~Reporting initial MFC integrated flow
2018-10-17 290 17:42:48:819~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*00g31)"
2018-10-17 290 17:42:49:010~acme.flow~Reply for command: *00g31
2018-10-17 290 17:42:49:010~acme.flow~"0
2018-10-17 290 17:42:49:010~acme.flow~>"
2018-10-17 290 17:42:49:819~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*01g31)"
2018-10-17 290 17:42:50:011~acme.flow~Reply for command: *01g31
2018-10-17 290 17:42:50:011~acme.flow~"0
2018-10-17 290 17:42:50:011~acme.flow~>"
2018-10-17 290 17:42:50:819~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31)"
2018-10-17 290 17:42:51:010~acme.flow~Reply for command: *02g31
2018-10-17 290 17:42:51:010~acme.flow~"0
2018-10-17 290 17:42:51:010~acme.flow~>"
2018-10-17 290 17:42:51:819~zt.script~Test point PAUSED to check igniter.
SystemResumeScript when ready.
2018-10-17 290 17:42:52:820~acme~"Received command: SystemPauseScript()"
2018-10-17 290 17:42:52:820~zt.script~Script paused.
2018-10-17 290 17:42:59:511~acme~"Received command: SystemResumeScript()"
2018-10-17 290 17:42:59:511~zt.script~Script resumed.
2018-10-17 290 17:42:59:523~zt.script~Configuring CIR manifolds, except fuel
2018-10-17 290 17:42:59:523~acme.flow~"Received command: FlowCloseAll()"

2018-10-17 290 17:42:59:524~acme~Sent IopRunScript command to IOP.
2018-10-17 290 17:42:59:524~zt.fcf~Sending out a command on the line.
2018-10-17 290 17:43:12:262~acme~"Received command:
SystemCirFlowState(fuelManifoldState:0,diluentManifoldState:0,oxygenManifoldState:0
,nitrogenManifoldState:2)"
2018-10-17 290 17:43:12:334~zt.script~CIR Manifold State. Fuel:Closed Diluent:Closed
Oxygen:Closed Nitrogen:Bypass
2018-10-17 290 17:43:12:334~acme.flow~"Received command:
FlowSetSolenoidValveState(solenoidValve:7,state:1)"
2018-10-17 290 17:43:12:835~zt.script~Priming nitrogen line for 10 sec
2018-10-17 290 17:43:12:835~acme.flow~"Received command:
FlowSetMfcRate(massFlowController:2,flowSlm:0.00723485)"
2018-10-17 290 17:43:22:837~zt.script~Recording pre-ignition data for 0 sec
2018-10-17 290 17:43:22:837~zt.script~Starting IPSU(s) archiving. Waiting up to 40 sec
2018-10-17 290 17:43:22:837~acme.diag~"Received command: DataRecordingStart()"
2018-10-17 290 17:43:22:837~acme.image~"Received command:
IpsuSetArchiveRate(ipsu:1,archiveRate:2)"
2018-10-17 290 17:43:22:838~acme.image~"Received command:
StartIpsus(testpointName:18290F1,ipsu5NumberCameras:2,ipsu5Camera1:3,ipsu5Came
ra2:4,ipsu5Camera3:0,ipsu5Camera4:0,ipsu6NumberCameras:0,ipsu6Camera1:0,ipsu6C
amera2:0,ipsu6Camera3:0,ipsu6Camera4:0)"
2018-10-17 290 17:43:22:839~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:43:22:841~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:43:22:847~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:43:22:848~acme.ipsu~Success for IPSU 5 command IpsugTestpointOpen
2018-10-17 290 17:43:22:848~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:43:22:850~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:43:22:852~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:43:22:853~acme.ipsu~Success for IPSU 5 command
IpsugCamerasStartTestPointArchive
2018-10-17 290 17:43:23:837~zt.script~Recording pre-ignition images for 0 sec
2018-10-17 290 17:43:23:837~zt.script~Opening CIR fuel manifold
2018-10-17 290 17:43:23:837~acme~Sent IopRunScript command to IOP.
2018-10-17 290 17:43:23:838~zt.fcf~Sending out a command on the line.
2018-10-17 290 17:43:38:923~acme~"Received command:
SystemCirFlowState(fuelManifoldState:1,diluentManifoldState:0,oxygenManifoldState:0
,nitrogenManifoldState:2)"
2018-10-17 290 17:43:38:951~zt.script~CIR Manifold State. Fuel:Open Diluent:Closed
Oxygen:Closed Nitrogen:Bypass
2018-10-17 290 17:43:38:951~zt.script~Intensified camera to continuous mode
2018-10-17 290 17:43:38:951~acme.image~"Received command:
IlluminationConfigure(sideLEDBankState:0,endLEDBankState:0)"
2018-10-17 290 17:43:38:952~acme.image~"Received command:
IntensifiedCameraGate(gateType:1,pulsedGateWidthCounts:0)"
2018-10-17 290 17:43:38:952~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:43:38:954~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:43:38:956~zt.fcf~Success sending RVC command in rvcThread

2018-10-17 290 17:43:39:451~acme.image~"Received command: IntensifiedCameraGainAmplitude(gainAmplitudeCounts:2643)"
2018-10-17 290 17:43:39:452~acme.igniter~"Received command: IgnitionDetectionSetState(state:1)"
2018-10-17 290 17:43:39:452~acme.igniter~Ignition detection enabled
2018-10-17 290 17:43:39:452~acme.igniter~"Received command: IgniterSetupMove(position:0)"
2018-10-17 290 17:43:39:594~acme.flow~"Received command: FlowSetAllSolenoidValveStates(SVA1State:0,SVA2State:0,SVA3State:0,SVA4State:0,SVA5State:0,SVA6State:0,SVA7State:1,SVA8State:0,SVA9State:1)"
2018-10-17 290 17:43:39:644~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:43:39:644~zt.fcf~Waiting on RVC response in rvcThreadSuccess sending RVC command in rvcThread
2018-10-17 290 17:43:39:951~zt.script~Flow timeout in 150 sec
2018-10-17 290 17:43:39:952~zt.script~Priming gas lines for 5 sec
2018-10-17 290 17:43:39:952~acme.flow~"Received command: FlowSetAllMfcRates(MFCA1StaticMixerFlowSlm:0,MFCA2BypassFlowSlm:0.00723485,MFCA3FuelFlowSlm:0.0477391)"
2018-10-17 290 17:43:44:953~zt.script~Igniting with 3.75V for 2000 Ms
2018-10-17 290 17:43:44:953~acme.igniter~"Received command: IgniterPulse(power:3.75,durationMs:2000)"
2018-10-17 290 17:43:45:752~acme.igniter~IGNITION DETECTED
2018-10-17 290 17:43:45:752~acme.igniter~Powering off igniter and homing
2018-10-17 290 17:43:46:453~zt.script~Changing cameras to test settings
2018-10-17 290 17:43:46:453~zt.script~Starting bracketed exposures on ACME camera
2018-10-17 290 17:43:46:453~zt.script~Waiting stabilization time #1 1.5 sec
2018-10-17 290 17:43:46:493~acme.igniter~"Received command: IgniterExecuteSetupMove()"
2018-10-17 290 17:43:46:633~acmeIgniterMotion.trace~void IgniterController::startMotion() local _inMotion already true.
2018-10-17 290 17:43:46:633~acmeIgniterMotion.trace~void IgniterController::startMotion() Motion timer already active!
2018-10-17 290 17:43:46:634~acme.image~"Received command: DataCameraExposureBracketing(enable:1,imagesPerExposure:1,exposures:6,exposureStartUsec:75,exposureIncrementUsec:340)"
2018-10-17 290 17:43:46:634~acme.image~"Received command: AnalogCameraSetShutter(shutter:5)"
2018-10-17 290 17:43:46:670~acme.image~Analog command successful.
2018-10-17 290 17:43:46:761~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:43:46:761~zt.fcf~Waiting on RVC response in rvcThreadSuccess sending RVC command in rvcThread
2018-10-17 290 17:43:47:453~zt.script~Enabling and initializing E-Field to -800 volts
2018-10-17 290 17:43:47:454~zt.script~Waiting E-Field initialize time 2 sec
2018-10-17 290 17:43:47:454~acme.efield~"Received command: EFieldEnable(state:1)"
2018-10-17 290 17:43:47:454~acme.efield~"Received command: EFieldSetVoltage(voltage:-800)"
2018-10-17 290 17:43:49:454~zt.script~Extinction detection enabled

2018-10-17 290 17:43:49:454~zt.script~Waiting for dilution time 0 secs
2018-10-17 290 17:43:49:454~acme.diag~"Received command: ExtinctionDetectionSetState(state:1)"
2018-10-17 290 17:43:49:454~zt.script~Waiting stabilization time #2 0 sec
2018-10-17 290 17:43:49:454~zt.script~Waiting for coflow to match test conditions 0 sec
2018-10-17 290 17:43:49:455~zt.script~Waiting stabilization time #3 0 sec
2018-10-17 290 17:43:49:455~zt.script~Waiting constant flow time 5 sec
2018-10-17 290 17:43:54:456~zt.script~Stepping E-Field
2018-10-17 290 17:43:54:456~acme.efield~"Received command: EFieldSteppedSignal(minVoltage:-10000,maxVoltage:-1000,stepVoltage:-350,stepDurationSecs:2,cycles:1)"
2018-10-17 290 17:44:45:463~zt.script~Waveform complete. E-Field disabled
2018-10-17 290 17:44:45:463~zt.script~Waiting constant flow time #2 5 sec
2018-10-17 290 17:44:45:494~acme.efield~"Received command: EFieldEnable(state:0)"
2018-10-17 290 17:44:50:465~zt.script~Turning off gas flow
2018-10-17 290 17:44:50:465~zt.script~Waiting post-flame imaging duration of 3 sec
2018-10-17 290 17:44:50:493~acme.igniter~"Received command: IgnitionDetectionSetState(state:0)"
2018-10-17 290 17:44:50:493~acme.igniter~Ignition detection disabled
2018-10-17 290 17:44:50:494~acme.diag~"Received command: ExtinctionDetectionSetState(state:0)"
2018-10-17 290 17:44:50:494~acme.flow~"Received command: FlowCloseAll()"
2018-10-17 290 17:44:50:495~acme~Sent IopRunScript command to IOP.
2018-10-17 290 17:44:50:496~zt.fcf~Sending out a command on the line.
2018-10-17 290 17:44:53:466~zt.script~Stopping IPSU(s)
2018-10-17 290 17:44:53:466~zt.script~Waiting post-flame data duration of 5 sec
2018-10-17 290 17:44:53:495~acme.image~"Received command: StopIpsus()"
2018-10-17 290 17:44:53:496~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:44:53:497~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:44:53:653~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:44:53:654~acme.ipsu~Success for IPSU 5 command IpsugCamerasStopTestPointArchive
2018-10-17 290 17:44:54:532~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:44:54:532~zt.fcf~Waiting on RVC response in rvcThreadSuccess sending RVC command in rvcThread
2018-10-17 290 17:44:54:532~acme.ipsu~Success for IPSU 5 command IpsugTestpointClose
2018-10-17 290 17:44:58:467~zt.script~Test point PAUSED prior to leak check measurement. SystemResumeScript when ready.
2018-10-17 290 17:44:58:494~acme.diag~"Received command: DataRecordingStop()"
2018-10-17 290 17:44:59:495~acme~"Received command: SystemPauseScript()"
2018-10-17 290 17:44:59:495~zt.script~Script paused.
2018-10-17 290 17:45:06:595~acme~"Received command: SystemResumeScript()"
2018-10-17 290 17:45:06:595~zt.script~Script resumed.
2018-10-17 290 17:45:06:671~acme.diag~"Received command: DataRecordingStart()"
2018-10-17 290 17:45:07:671~zt.script~Measuring leak current
2018-10-17 290 17:45:07:671~zt.script~E-field enabled
2018-10-17 290 17:45:07:671~zt.script~Initializing E-Field to -800 volts

2018-10-17 290 17:45:07:671~acme.efield~"Received command: EFieldEnable(state:1)"
2018-10-17 290 17:45:07:671~zt.script~Waiting E-Field initialize time 2 sec
2018-10-17 290 17:45:07:672~acme.efield~"Received command: EFieldSetVoltage(voltage:-800)"
2018-10-17 290 17:45:09:672~zt.script~Stepping E-Field
2018-10-17 290 17:45:09:672~acme.efield~"Received command: EFieldSteppedSignal(minVoltage:-10000,maxVoltage:-1000,stepVoltage:-350,stepDurationSecs:2,cycles:1)"
2018-10-17 290 17:46:00:680~zt.script~Waveform complete. E-Field disabled
2018-10-17 290 17:46:00:680~acme.efield~"Received command: EFieldEnable(state:0)"
2018-10-17 290 17:46:06:680~zt.script~Intensified camera to safe mode
2018-10-17 290 17:46:06:680~acme~"Received command: SystemCloseTestPoint()"
2018-10-17 290 17:46:06:680~acme~List of files for test point "18290F1"
2018-10-17 290 17:46:06:681~acme~"/payload/sci/18290F1/18290F1_data.csv"
2018-10-17 290 17:46:06:681~acme~"/payload/sci/18290F1/18290F1_mfcSetpoints.csv"
2018-10-17 290 17:46:06:726~acme~"/payload/logs/2018-10-17_12-26ACME.log"
2018-10-17 290 17:46:06:726~acme~"/payload/logs/2018-10-17_12-59ACMECommands.log"
2018-10-17 290 17:46:06:726~acme.image~"Received command: IntensifiedCameraGate(gateType:0,pulsedGateWidthCounts:0)"
2018-10-17 290 17:46:06:727~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:46:06:737~zt.fcf~Waiting on RVC response in rvcThreadSuccess sending RVC command in rvcThread
2018-10-17 290 17:46:06:927~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:46:06:929~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:46:07:144~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:46:07:181~zt.script~Setting PMT gains to zero
2018-10-17 290 17:46:07:181~acme.image~"Received command: IntensifiedCameraGainAmplitude(gainAmplitudeCounts:2048)"
2018-10-17 290 17:46:07:181~zt.script~Reporting final MFC integrated flow
2018-10-17 290 17:46:07:181~acme.image~"Received command: DataCameraExposureBracketing(enable:0,imagesPerExposure:1,exposures:6,exposureStartUsec:75,exposureIncrementUsec:340)"
2018-10-17 290 17:46:07:182~acme.diag~"Received command: PmtSetAllGains(PMT1Gain:0,PMT2Gain:0,PMT3Gain:0)"
2018-10-17 290 17:46:07:183~acme.flow~"Received command: FlowMfcSerialCommand(serialCommand:*00g31)"
2018-10-17 290 17:46:07:183~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:46:07:183~zt.fcf~Sending out an RVC command on the line in rvcThread.
2018-10-17 290 17:46:07:185~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:46:07:187~zt.fcf~Waiting on RVC response in rvcThread
2018-10-17 290 17:46:07:231~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:46:07:233~zt.fcf~Success sending RVC command in rvcThread
2018-10-17 290 17:46:07:514~acme.flow~Reply for command: *00g31
2018-10-17 290 17:46:07:514~acme.flow~"0
2018-10-17 290 17:46:07:514~acme.flow~>"
2018-10-17 290 17:46:08:181~acme.flow~"Received command:

FlowMfcSerialCommand(serialCommand:*01g31)"
2018-10-17 290 17:46:08:526~acme.flow~Reply for command: *01g31
2018-10-17 290 17:46:08:526~acme.flow~"0.01123948
2018-10-17 290 17:46:08:526~acme.flow~>"
2018-10-17 290 17:46:09:181~acme.flow~"Received command:
FlowMfcSerialCommand(serialCommand:*02g31)"
2018-10-17 290 17:46:09:525~acme.flow~Reply for command: *02g31
2018-10-17 290 17:46:09:525~acme.flow~"0.05557147
2018-10-17 290 17:46:09:525~acme.flow~>"
2018-10-17 290 17:46:10:181~zt.script~E-Field Voltage Sweep test point 18290F1
COMPLETE
2018-10-17 290 17:46:21:792~acme.image~"Received command:
IlluminationConfigure(sideLEDBankState:1,endLEDBankState:1)"

IPSUG.log excerpt

Meanwhile, the IPSUG.log files can provide a check of the exposure time, which can be particularly important when bracketing is employed with the ACME DataCam. The corresponding excerpt for test 18290F1, in which bracketing is used, is presented here without the many repetitive rows in the middle.

2018-10-17 290 17:42:28:206~ipsug~CPU operational minutes is now 137855
2018-10-17 290 17:42:43:433~ipsug.Halcon~SetExposure(integrationTimeUsecs 100000)
2018-10-17 290 17:43:22:852~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1"
2018-10-17 290 17:43:22:852~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1/sci"
2018-10-17 290 17:43:22:852~zt.filesys~Set new directory permissions to rwxrwxrwx (777)
for: "/sd1/sci/18290F1"
2018-10-17 290 17:43:22:856~ipsug.image~virtual void
ImagingController::handleCommand(XGenericMessage*) bCmdComplete camera 3
2018-10-17 290 17:43:22:857~ipsug.image~virtual void
ImagingController::handleCommand(XGenericMessage*) bCmdComplete camera 4
2018-10-17 290 17:43:22:857~ipsug.FGrabber~void HalconFrameGrabber::captureImages()
"ACME" start of capture
2018-10-17 290 17:43:22:857~ipsug.FGrabber~void HalconFrameGrabber::captureImages()
"INTNS" start of capture
2018-10-17 290 17:43:28:188~ipsug~CPU operational minutes is now 137856
2018-10-17 290 17:43:46:679~ipsug.Halcon~setExposureBracketing(1 , 6 , 75 , 340)
2018-10-17 290 17:43:46:680~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:43:46:711~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:43:46:812~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:43:46:850~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:43:46:880~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:43:46:915~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
2018-10-17 290 17:43:46:947~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:43:46:981~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:43:47:014~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)

2018-10-17 290 17:43:47:050~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:43:47:081~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:43:47:115~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
2018-10-17 290 17:43:47:145~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:43:47:179~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:43:47:217~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:43:47:248~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:43:47:282~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:43:47:313~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
.
.
.
2018-10-17 290 17:44:52:860~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:44:52:899~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:44:52:932~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:44:52:967~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:44:53:000~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:44:53:163~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
2018-10-17 290 17:44:53:169~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:44:53:175~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:44:53:180~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:44:53:187~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:44:53:193~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:44:53:228~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
2018-10-17 290 17:44:53:261~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:44:53:296~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:44:53:333~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:44:53:365~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:44:53:397~ipsug.Halcon~SetExposure(integrationTimeUsecs 10019)
2018-10-17 290 17:44:53:431~ipsug.Halcon~SetExposure(integrationTimeUsecs 34064)
2018-10-17 290 17:44:53:465~ipsug.Halcon~SetExposure(integrationTimeUsecs 75)
2018-10-17 290 17:44:53:496~ipsug.Halcon~SetExposure(integrationTimeUsecs 255)
2018-10-17 290 17:44:53:532~ipsug.Halcon~SetExposure(integrationTimeUsecs 867)
2018-10-17 290 17:44:53:550~ipsug.FGrabber~"INTNS" saving meta data after archiving
2018-10-17 290 17:44:53:557~ipsug.FGrabber~"ACME" saving meta data after archiving
2018-10-17 290 17:44:53:558~ipsug.Halcon~SetExposure(integrationTimeUsecs 2947)
2018-10-17 290 17:44:53:610~ipsug.FGrabber~"ACME" end of capture in captureImages
2018-10-17 290 17:44:53:611~ipsug.FGrabber~"INTNS" end of capture in captureImages
2018-10-17 290 17:45:28:196~ipsug~CPU operational minutes is now 137858
2018-10-17 290 17:46:07:200~ipsug.Halcon~setExposureBracketing(0 , 0 , 0 , 0)

Appendix A
Nomenclature

This document's key abbreviations are listed here.

ACME	Advanced Combustion via Microgravity Experiments (project including E-FIELD Flames)
AOI	Area of Interest (e.g., for image acquisition of just a portion of the camera FOV)
BRE	Burning Rate Emulator (another ACME experiment)
CCD	Charge-Coupled Device (such as a camera sensor)
CIR	Combustion Integrated Rack (ISS facility in which ACME experiments are conducted)
CFI	Cool Flames Investigation (a droplet combustion experiment preceding ACME in the CIR)
CLD Flame	Coflow Laminar Diffusion Flame (another ACME experiment)
CSV	comma-separated values
DataCam	Data Camera (for ACME experiments, characterized for pyrometry, etc.)
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames
Fiji	Fiji Is Just ImageJ (a version of ImageJ preloaded with many plug-in features)
FOMA	Fuel Oxidizer Management Assembly (CIR's gas management subsystem)
FOV	Field of View
fps	frames per second
GCIP	Gas Chromatograph Instrumentation Package
GIPOC	Glenn ISS Payload Operations Center
GMT	Greenwich Mean Time
HiBMs	High Bit-depth Multi-spectral camera (of the CIR facility)
HOBJ	Halcon object (binary uncompressed format; used for the DataCam)
ImageJ	Image Processing and Analysis in Java (open-source image processing software)
IPSU	Image Processing and Storage Unit (CIR subsystem)
IPSU-G	IPSU for Gigabit ethernet cameras
ISS	International Space Station
LLL-UV	Low Light Level Ultraviolet camera (of the CIR facility)
MFC	Mass Flow Controller
NTSC	National Television System Committee
OMA2	Optical Multi-channel Analyzer 2 (open-source software for image processing and display)
ops	operations
PaRIS	Passive Rack Isolation System
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (where ACME is equipped with three PMT modules)
PT	pressure transducer
psia	pounds per square inch
s-Flame	Structure and Response of Spherical Diffusion Flame (another ACME experiment)
SAMS	Space Acceleration Measurement System
slpm	standard liters per minute (a measure of the mass or molar flow rate of a gas)
TFP	Thin-Filament Pyrometry (a technique for making gas-phase temperature measurements)
TIFF or TIF	Tagged Image File Format (used by the CFI, HiBMs, and LLL-UV cameras)
TM	thermistor
TS	Transport Stream (a video format)

Appendix B
Installation of QtOMA2 on a PC

These instructions were provided by Prof. Marshall Long (Yale U.).

These instructions assume you don't already have Qt installed. If you do, skip to Part B.

Part A:

Download the opensource Qt from https://www.qt.io/download.

- ➔ Go open source
- ➔ Download the Qt Online Installer
- ➔ Download
- ➔ Save File

In the downloads folder, open qt-unified-windows-x86-x.x.x-online

- ➔ Sign up for a Qt account or login
- ➔ Accept Open Source obligations
- ➔ Contribute to Qt Development (on not)
- ➔ Accept default install location
- ➔ Install the components as shown in the figure on the next page
- ➔ Agree to terms
- ➔ Add shortcuts
- ➔ Install

Part B:

Download the latest version of QtOma2 from https://github.com/mblong/qtoma2.

- ➔ Code → Download ZIP

Expand the zip file.
In the qtoma2-master folder, open QtOma.pro (the Qt Project file)

The first time this opens in Qt Creator, it will ask for the project to be configured. Just choose the defaults.

- ➔ Configure Project
- ➔ In the lower left corner, click on the green arrow to run QtOma2.

Selection of Qt (not OMA2) components to install in part A

